UNIVERSITY OF WOLLONGONG AUSTRALIA

SIM GLOBAL EDUCATION

CSIT-21-S2-15 : MY APPOINTMENT

FYP-21-S2-15 User Requirement Specification

Team

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Project Website

https://fyp565230019.wordpress.com

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1. Introduction

1.1 Purpose

The purpose of this user requirement specification document is to help assist the team in the approval of the My Appointment project. This document will aid us in the decision of the requirements of our project. Also, the features of each user will be implemented with the requirements that we have decided on.

1.2 Intended Audience and Reading Suggestions

This user requirement specification document intended audience is for the people who will be involved in this project. This includes us, the people who are working on the project, along with both our supervisor, Mr Premrajan P and our assessor, Mr Tian Sion Hui. Mandatory knowledge with regards to Information Technology is required to be able to read through this document.

1.3 Project Scope

The scope of this project is to provide patients an online system to manage their appointments. Also, there will be three different tiers of subscriptions and is available for website on both computer and mobile.

2. Overall Description

2.1 Product Perspective

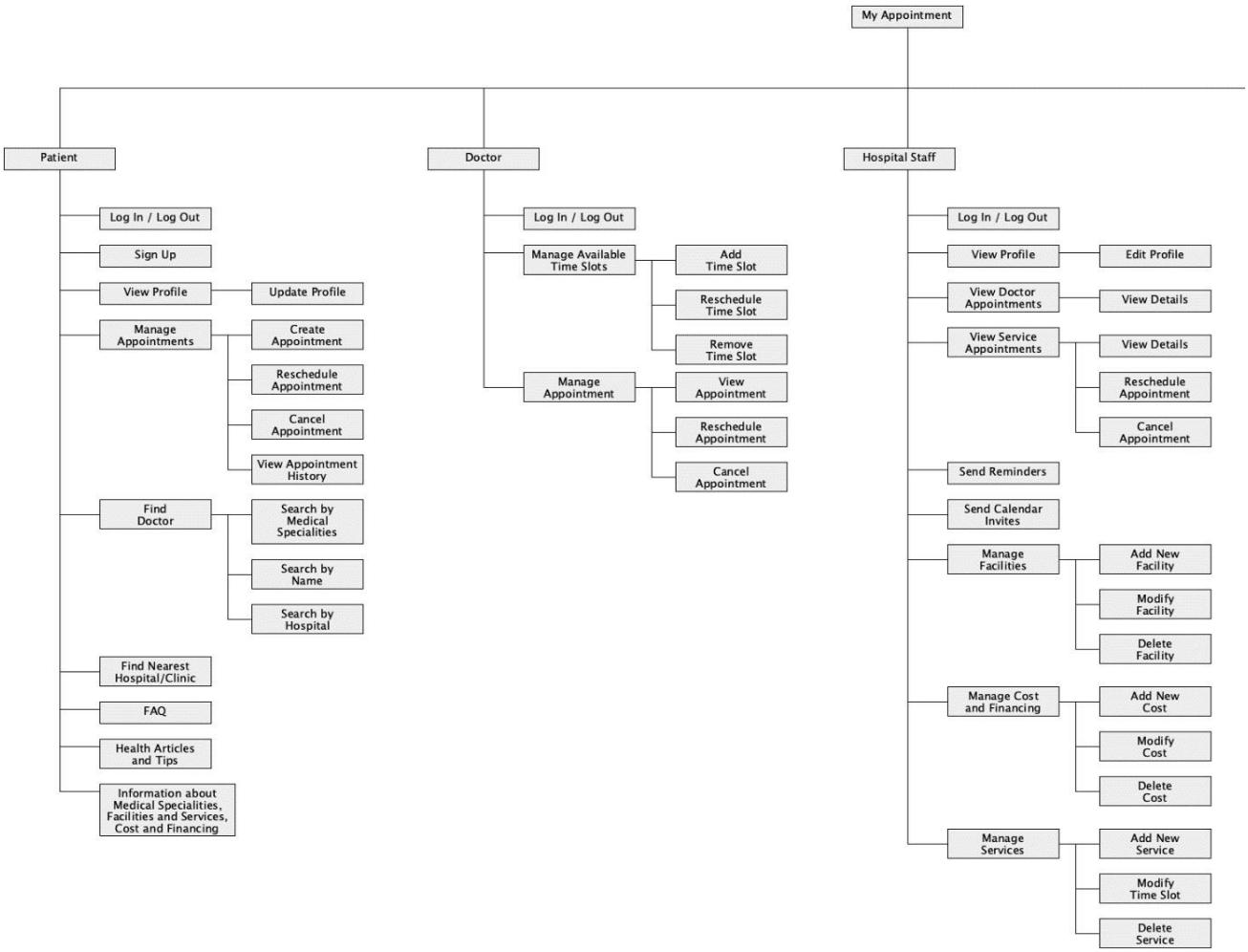
The product we are developing is a website application. This website application will be available for patients, hospitals, and clinics. The website allows the patients to book and reschedule medical appointments. It also allows the patients to view the facilities, services, and medical specialties provided by the hospitals and/or clinics. Hospitals and clinics can manage the appointments made by the patients, also send reminders and calendar invites to the patients.

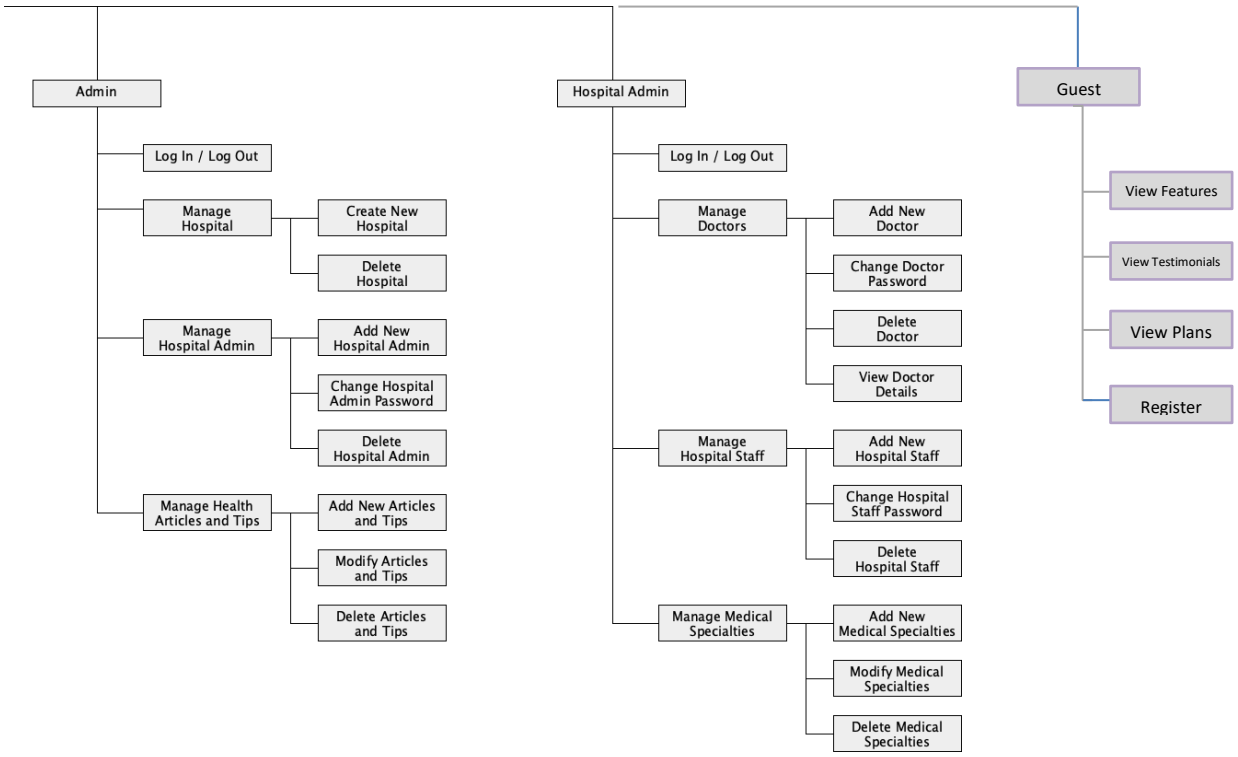
The patients have to register for an account to log on into our website to make or reschedule their appointments. The website also includes features like find a doctor, health articles and tips, cost and financing, and FAQ. The website also provides a secure way of storage and transfer of patients' sensitive data.

For the hospital staff, they are able to view and manage the appointments made by the patients, send reminders, send calendar invites, manage doctors information, and manage hospital's information.

The admin is able to create hospital staff accounts, change the password for hospital staff accounts, and also delete the hospital staff accounts. Admin also manages the health articles, health tips, and the FAQ.

2.2 Product Features




Admin
Log In / Log Out
Manage Hospital
Create New Hospital
Delete Hospital
Manage Hospital Admin
Add New Hospital Admin
Change Hospital Admin Password
Delete Hospital Admin
Manage Health Articles and Tips
Add New Articles and Tips
Modify Articles and Tips
Delete Articles and Tips
Hospital Admin
Log In / Log Out
Manage Doctors
Add New Doctor
Change Doctor Password
Delete Doctor
View Doctor Details
Manage Hospital Staff
Add New Hospital Staff
Change Hospital Staff Password
Delete Hospital Staff
Manage Medical Specialties
Add New Medical Specialties
Modify Medical Specialties
Delete Medical Specialties
Guest
View Features
View Testimonials
View Plans
Register

Patient	
FEATURE-1	Log In / Log Out
FEATURE-2	Sign Up / Register
FEATURE-3	View Profile
FEATURE-3.1	Update Profile
FEATURE-4	Manage Appointments
FEATURE-4.1	Create Appointment
FEATURE-4.2	Reschedule Appointment
FEATURE-4.3	Cancel Appointment
FEATURE-4.4	View Appointment History
FEATURE-5	Find Doctor
FEATURE-5.1	Search by Medical Services
FEATURE-5.2	Search by Name
FEATURE-5.3	Search by Hospital
FEATURE-6	Find Nearest Hospital / Clinic
FEATURE-7	Health Articles and Tips
FEATURE-8	Access Information about Medical Specialties, Facilities and Services, Cost and Financing
FEATURE-9	FAQ

Doctor	
FEATURE-1	Log In / Log Out
FEATURE-2	Manage Appointments
FEATURE-2.1	Cancel Appointments
FEATURE-2.2	Reschedule Appointments
FEATURE-2.3	View Appointment Details
FEATURE-3	Manage Time Slots
FEATURE-3.1	Add Time Slots
FEATURE -3.2	Reschedule Time Slots
FEATURE -3.3	Remove Time Slots

Hospital Staff	
FEATURE-1	Log In / Log Out
FEATURE-2	Update Account Information
FEATURE-3	Manage Services
FEATURE-3.1	Manage Services Timeslots
FEATURE-3.2	Add New Service
FEATURE-3.3	Delete Service
FEATURE-4	View Doctor Appointments
FEATURE-5	View Service Appointments
FEATURE-5.1	Reschedule Service Appointments
FEATURE-5.2	Cancel Service Appointments
FEATURE-6	Send Reminders
FEATURE-7	Send Calendar Invites
FEATURE-8	Manage Information about Facilities and Services, Cost and Financing

Hospital Admin	
FEATURE-1	Log In / Log Out
FEATURE-2	Create Doctor Account
FEATURE-3	Delete Doctor Account
FEATURE-4	Create Hospital Staff Account
FEATURE-5	Delete Hospital Staff Account
FEATURE-6	Change Doctor and Hospital Staff Password
FEATURE-7	View Medical Specialties
FEATURE-7.1	Add Medical Specialties
FEATURE-7.2	Delete Medical Specialties
FEATURE-7.3	Modify Medical Specialties

Admin	
FEATURE-1	Log In / Log Out
FEATURE-2	Create Hospital
FEATURE-3	Delete Hospital
FEATURE-4	Create Hospital Admin Account
FEATURE-5	Delete Hospital Admin Account
FEATURE-6	Manage Health Articles and Tips
FEATURE-7	Manage FAQ

2.3 User Classes and Characteristics

Users	Characteristics
Patient	The patient will be able to log in and out of the system, sign up to create an account at the website. Once the patient is registered, they can view and update their profile as required. They can even manage their appointments and our system allows them to create, reschedule, cancel their appointments, and also allows them to view their appointment history. Our system also aids the patients to find a doctor and offers them different search criteria such as 'Search by Medical Specialities', 'Search by Name' and 'Search by Hospital'. The patients can also find the nearest hospital/clinic their residence by entering their postal code following which the system would generate a list of nearby healthcare institutions. The system has also planned on including FAQ, Health Articles and Tips, Information about Medical Specialities, Facilities and Services, Cost and Financing for the patients so that they can gather essential information in a few clicks.
Doctor	The doctor can log in and out of the system. They can manage their available time slots. The doctor can even manage the appointments that his patients had booked. For instance, he can view the appointment and know before-hand the complications/disease the patient has. Next, the doctor can even request the patient to reschedule the appointment to a different day or timing. Lastly, the doctor can even cancel the appointment in the case where the doctor is himself/herself not keeping well or has an urgent task to handle.

Hospital Staff	Hospital Staff can log in and out of the system, managing the service appointments (view details, reschedule, and cancel). The hospital staff can also view the doctor appointments. The are also able to send reminders and calendar invites. The hospital staff is also responsible for managing the facilities and services in the hospital.
Hospital Admin	Hospital Admin is responsible for creating and managing doctor and hospital staff account. They are also responsible for creating and managing the medical specialties in the hospital.
Admin	Admin can log in and out of the system, they can also manage the health articles and health tips that will be shown in the patient's home page. Admin also manages the FAQ for the system. Other than that, admin also create and manage the hospital and hospital admin.

2.4 Operating Environment

Operating Environment 1	The website application requires a browser that is compatible with html, CSS, JavaScript, and php
Operating Environment 2	The website application requires access to internet connection
Operating Environment 3	All personal data should be encrypted

2.5 Design and Implementation Constraints

Constraint 1	UI should be created through HTML, CSS, JavaScript, PHP, and MySQL.
Constraint 2	Data should be stored in MySQL Workbench.
Constraint 3	The website application requires access to a stable internet connection.
Constraint 4	For the best experience we recommend the Internet Explorer 11 or Mozilla Firefox 45 or Chrome 48.
Constraint 5	A secure database will be used to store the data of Patients, Hospital Staff, and Admin.

2.6 User Documentation

Upon entering the webpage, users will be required to enter their login credentials. After logging in, users will be shown their upcoming appointment if any, else users will be shown upcoming events the hospital has hosted. Users will be able to express interest in participating in the event. The website allows the users to book an appointment with the doctors in the respective category they desire with the help of the search function. The website also offers health tips and FAQs to answer any questions the users might have.

2.7 Assumption and Dependencies

Assumption 1	In order to have the best experience of using the web application we recommend our users to use either of the following Internet Explorer 11 or Mozilla Firefox 45 or Chrome 48.
Assumption 2	The web application demands for a stable internet connection.
Assumption 3	The web application also assumes that your Workstation is able to load the html files.
Assumption 4	The web application assumes that its users enter genuine information for proper functioning of the web application and prevent errors from arising.

3. System Features

3.1 Use Case Diagram - Patient

3.1.1 User Stories - Patient

- As a patient I want to be able to log in so that I can access the system
- As a patient I want to be able to log out so that I can exit the system
- As a patient I want to be able to sign up / register so that I can access the system
- As a patient I want to be able to view my personal information so that I can check on my information
- As a patient I want to be able to update or modify my personal information so that my account information is up to date
- As a patient I want to be able to create an appointment so that I can get a schedule to meet with doctors
- As a patient I want to be able to change an appointment date and time so that I can shift my appointment if I'm unable to attend
- As a patient I want to be able to cancel an appointment so that I can notify the hospital that I cancel my appointment
- As a patient I want to be able to view appointments so that I can take notice of any upcoming appointments
- As a patient I want to be able to search for a doctor information so that I can choose a medical service which is rendered by the specific doctor.
- As a patient I want to be able to search for the nearest hospital and clinics so that I can find hospitals that are close to me
- As a patient I want to be able to view the FAQ so that I can understand the system better
- As a patient I want to be able to access health articles & health tips so that I can be up to date with the newest and latest health information.
- As a patient I want to be able to access my appointments history that are kept in the database so that I can view my appointment history.
- As a patient I want to be able to access information about the medical specialties, facilities, services, and cost so that I can better understand the services that I choose and the details of the services as well.
- As a patient I want to be able to compare the costs of facilities at different hospitals so that I can find the best suitable rate for myself.

3.1.2 Use Case Description - Patient

Name: Log In (Patient)	**ID: P01**
Stakeholders and goals: Patient – Patient can successfully login into the system	
Description: A patient needs to login before using the system.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Login button	
Normal Flow: 1. **The patient will click on the Login button** 2. **System shows the login form** 3. **The patient will enter the login credentials** 4. **System will validate the patient** 5. **Patient will be logged in**	
Sub-Flows: None	
Alternative/Exceptional Flows: **4a. System is unable to authenticate user** **5a. Patient is unable to log in to the system**	

Name: Sign Up (Patient)	**ID: P02**
Stakeholders and goals: Patient – Patient can successfully register into the system	
Description: A patient needs to register before the patient can use the system.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Sign-Up button	
Normal Flow: 1. **The patient will click on the Sign-Up button** 2. **System shows the sign-up form** 3. **The patient will enter the details on the sign-up form** 4. **System will validate the information and save the information into the database** 5. **Patient will be registered**	
Sub-Flows: None	
Alternative/Exceptional Flows: **4a. Patient's information is invalid** **5a. Patient is unable to register to the system**	

Name: View Profile	ID: P03
Stakeholders and goals: Patient – Patient can successfully view their personal information	
Description: A patient should be able to view their personal information on the system.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the View Profile button	
Normal Flow: 1. The patient will click on the View Profile button 2. System shows the Profile Information page	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Update Profile	**ID: P04**
Stakeholders and goals: Patient – Patient can successfully update their personal information	
Description: A patient should be able to update their personal information on the system.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Update Profile button	
Normal Flow: 1. **The patient will click on the Update Profile button** 2. **System shows the Update Profile form** 3. **Patient enters the information they want to update** 4. **The patient will click on the Update button** 5. **System validates the updated information** 6. **Patient's Profile is updated**	
Sub-Flows: None	
Alternative/Exceptional Flows: **5a. Patient's information is not valid** **6a. Failed to update Patient's Profile**	

Name: Manage Appointment	ID: P05
Stakeholders and goals: Patient – Patient can successfully manage his/her appointment(s)	
Description: A patient should be able to manage their appointment(s).	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Manage Appointment button	
Normal Flow: 1. **The patient will click on the Manage Appointment button** 2. **System shows the Manage Appointment page which contains the list of appointment(s) and options to Create Appointment, Re-Schedule Appointment, Cancel Appointment, and View Appointment History**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Create Appointment	ID: P06
Stakeholders and goals: Patient – Patient can successfully create an appointment	
Description: Patient wants to create an appointment.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Create Appointment button	
Normal Flow: 1. The patient will click on the Create Appointment button 2. System shows the create appointment form 3. Patient enters appointment details 4. System validates the information entered by the patient 5. Appointment created	
Sub-Flows: None	
Alternative/Exceptional Flows: 4a. Information entered by the patient is invalid 5a. Failed to create an appointment	

Name: Reschedule Appointment	ID: P07
Stakeholders and goals: Patient – Patient can successfully re-schedule an appointment	
Description: Patient wants to reschedule an appointment.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Reschedule Appointment button	
Normal Flow: 1. The patient will click on the Reschedule Appointment button 2. System shows the reschedule appointment form 3. Patient enters a new date and time for the appointment 4. System validates the information entered by the patient 5. Appointment rescheduled	
Sub-Flows: None	
Alternative/Exceptional Flows: 4a. Information entered by the patient is invalid 5a. Failed to reschedule an appointment	

Name: Cancel Appointment	**ID: P08**
Stakeholders and goals: Patient – Patient can successfully cancel an appointment	
Description: Patient wants to cancel an appointment.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Cancel Appointment button	
Normal Flow: **1. The patient will click on the Cancel Appointment button** **2. System prompts the user for confirmation on the cancellation** **3. Patient confirms the cancellation** **4. Appointment cancelled**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a. Users cancel the appointment cancellation** **4a. Failed to cancel an appointment**	

Name: View Appointment History	ID: P09
Stakeholders and goals: Patient – Patient can successfully view the appointment history / records	
Description: Patient wants to view the appointment history.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the View Appointment History button	
Normal Flow: 1. The patient will click on the View Appointment History button 2. System will display the Appointment History page	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Find Doctor	ID: P10
Stakeholders and goals: Patient – Patient can search for doctor information	
Description: Patient should be able to search for doctor information by medical specialties, name, or hospital	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Find Doctor button	
Normal Flow: 1. The patient will click on the Find Doctor button 2. System will display a list of doctors and their information, including options to search doctors by specialty, name, or hospital	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Find Nearest Hospital	ID: P11
Stakeholders and goals: Patient – Patient can successfully search for nearest hospital / clinic	
Description: Patient should be able to search for nearest hospital / clinic.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Find Nearest Hospital button	
Normal Flow: 1. The patient will click on the Find Nearest Hospital button 2. System will display the find nearest hospital page 3. Patient enters a postal code 4. System validates the input 5. System displays the nearest hospital	
Sub-Flows: None	
Alternative/Exceptional Flows: 4a. Patient input is invalid 5a. System unable to display the nearest hospital	

Name: FAQ	ID: P12
Stakeholders and goals: Patient – Patient can successfully view the FAQ	
Description: Patient wants to view the FAQ.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the FAQ button	
Normal Flow: 1. The patient will click on the FAQ button 2. System will display the FAQ page	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

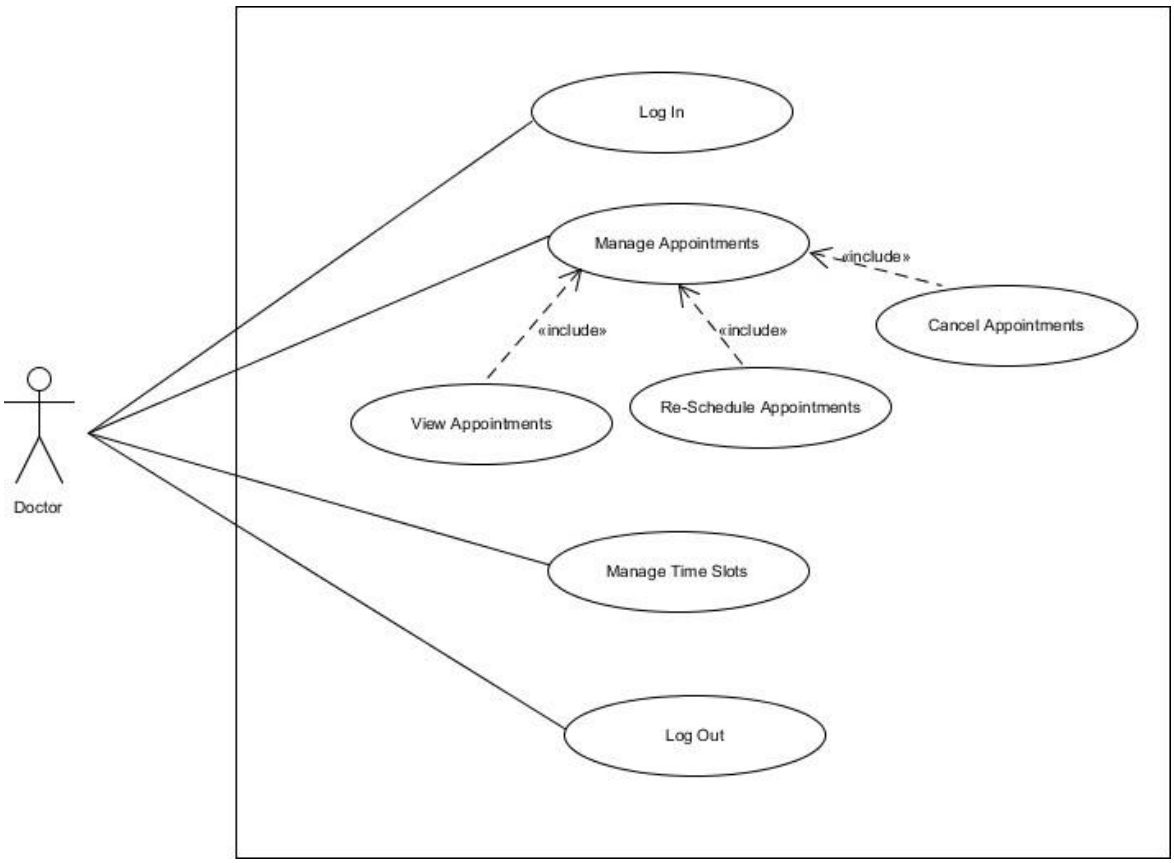
Name: Health Articles and Health Tips	ID: P13
Stakeholders and goals: Patient – Patient can successfully view the Health Articles and Health Tips	
Description: Patient wants to view the Health Articles and Health Tips.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Health Articles and Tips button	
Normal Flow: 1. **The patient will click on the Health Articles and Tips button** 2. **System will display the health articles and tips page**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Information about Medical Specialties, Facilities and Services, Cost and Financing	ID: P14
Stakeholders and goals: Patient – Patient can successfully view the information about a hospital	
Description: Patient wants to view information about a hospital.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Hospital Information button	
Normal Flow: 1. The patient will click on the Hospital Information button 2. Patient chooses a hospital 3. System will display the hospital information chosen by the patient	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Log Out	ID: P15
Stakeholders and goals: Patient – Patient can successfully log out from the system	
Description: Patient wants to log out from the system.	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Log Out button	
Normal Flow: 1. The patient will click on the Log Out button 2. Patient successfully logged out from the system	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Compare the costs of facilities at different hospitals	**ID: P16**
Stakeholders and goals: Patient – Patient can compare the costs of facilities at different hospitals	
Description: Patient wants to compare the costs of facilities at different hospitals	
Actors: Patient	
Trigger: This event will be triggered when the user clicks on the Compare button	
Normal Flow: 1. **The patient clicks on 'Compare Cost' button** 2. **The system displays the Compare Cost page** 3. **The patient selects two hospitals** 4. **The system displays the list of facilities with their relevant charges** 5. **The patient can now make a choice between the two hospitals**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

3.2 Use Case Diagram – Doctor



3.2.1 User Stories - Doctor

- As a doctor I want to be able to log in so that I can access the system
- As a doctor I want to be able to manage appointments so that I can view appointments
- As a doctor I want to be able to manage appointments so that I can cancel appointments
- As a doctor I want to be able to manage appointments so that I can reschedule appointments
- As a doctor I want to be able to manage time slots add time slots
- As a doctor I want to be able to manage time slots reschedule time slots
- As a doctor I want to be able to manage time slots remove time slots
- As a doctor I want to be able to log out so that I can exit the system

3.2.2 Use Case Description - Doctor

Name: Log In (Doctor)	**ID: D01**
Stakeholders and goals: Doctor – Doctor can successfully login into the system	
Description: Doctor needs to login before using the system.	
Actors: Doctor	
Trigger: This event will be triggered when the user clicks on the Login button	
Normal Flow: 1. **The Doctor will click on the Login button** 2. **System shows the login form** 3. **The Doctor will enter the login credentials** 4. **System will validate the Doctor** 5. **Doctor will be logged in**	
Sub-Flows: None	
Alternative/Exceptional Flows: **4a. System is unable to authenticate user** **5a. Doctor is unable to log in to the system**	

Name: Manage Appointments	ID: D02
Stakeholders and goals: Doctor – Doctor can manage their appointments	
Description: Doctor should be able to manage appointments.	
Actors: Doctor	
Trigger: This event will be triggered when the user clicks on the Manage Appointments button	
Normal Flow: 1. **The Doctor will click on the Manage Appointment button** 2. **System shows the list of appointments and options to reschedule and cancel**	
Sub-Flows: None	
Alternative/Exceptional Flows: **None**	

Name: Cancel Appointment	ID : D03
Stakeholders and goals: Doctor – Doctor can successfully cancel appointment	
Description: A Doctor should be able to cancel appointments.	
Actors: Doctor	
Trigger: This event will be triggered when the user clicks on the Cancel Appointment button	
Normal Flow: 1. **The Doctor will click on the Cancel Appointment button** 2. **System asks for confirmation on the cancellation** 3. **Doctor confirms the cancellation** 4. **Appointment cancelled**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a. Doctor cancel the cancellation** **4a. Failed to cancel the appointment**	

Name: Manage Time Slots	ID: D04
Stakeholders and goals: Doctor – Doctor can successfully manage time slots available for appointment	
Description: A Doctor should be able to manage time slots for appointments.	
Actors: Doctor	
Trigger: This event will be triggered when the user clicks on the Manage Time Slots button	
Normal Flow: 1. The Doctor will click on the Manage Time Slots button 2. Doctor manages and modifies the time slots 3. System validates the updated time slots 4. Time slots updated	
Sub-Flows: None	
Alternative/Exceptional Flows: 3a. Updated time slots are not valid 4a. Failed to update time slots	

Name: Add Time Slots	ID: D05
Stakeholders and goals: Doctor – Doctor can successfully add a time slots available for appointment	
Description: A Doctor should be able to add time slots for appointments.	
Actors: Doctor	
Trigger: This event will be triggered when the user clicks on the Add Time Slots button	
Normal Flow: 5. The Doctor will click on the Add Time Slots button 6. Doctor adds the time slots 7. System validates the newly input time slots 8. Time slots created	
Sub-Flows: None	
Alternative/Exceptional Flows: 3a. Added time slots are not valid 4a. Failed to add time slots	

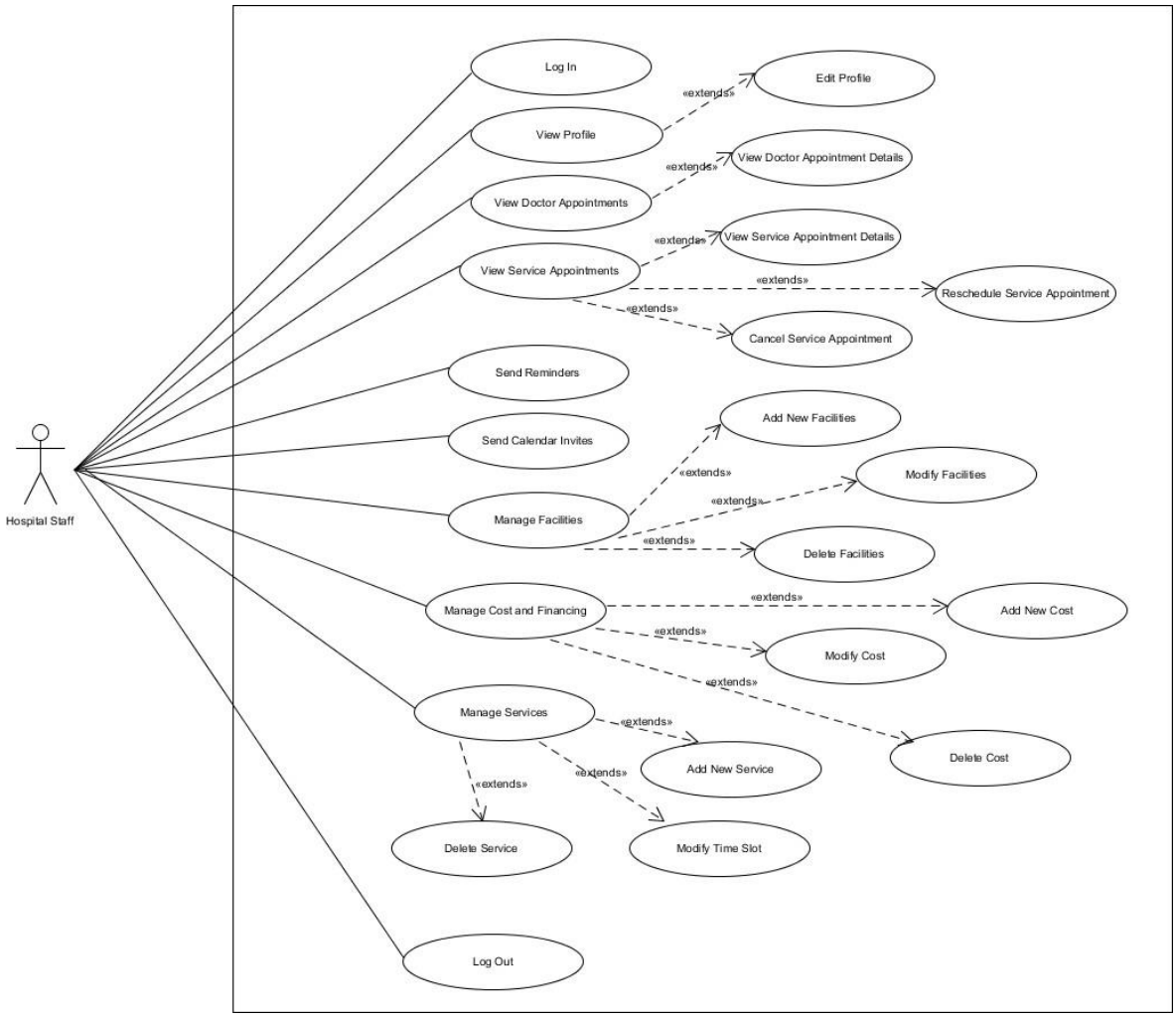
Name: Reschedule Appointment	ID: D06
Stakeholders and goals: Doctor – Doctor can successfully reschedule appointments made by the patients	
Description: Doctor wants to reschedule the appointments that are made by the patients.	
Actors: Doctor	
Trigger: This event will be triggered when the doctor clicks on the reschedule Appointment button	
Normal Flow: 1. The Doctor will click on the reschedule Appointments button 2. System will show the reschedule appointment form 3. Doctor enters a new date and time for the appointment 4. System validates the information entered by the doctor 5. Appointment Rescheduled	
Sub-Flows: None	
Alternative/Exceptional Flows: 4a. Information entered by the doctor is invalid 5a. Failed to reschedule an appointment	

Name: Reschedule Time Slots	ID: D07
Stakeholders and goals: Doctor – Doctor can successfully reschedule time slots available for booking	
Description: A doctor should be able to change the time slot made available for booking according to their availability	
Actors: Doctor	
Trigger: This event will be triggered when the doctor clicks on Manage Time Slots button	
Normal Flow: 1. The Doctor clicks on Manage Time Slots button 2. System will show all time slots available for the doctor to hold their appointment 3. Doctor will select the time slots that they are available for booking 4. Doctor confirms and system will update the available time slots accordingly	
Sub-Flows: None	
Alternative/Exceptional Flows: 4a. Time slot is not valid 4b. Time slot is no longer available	

Name: Remove Time Slots (Doctor)	ID : D08
Stakeholders and goals: Doctor – Doctor can successfully remove the time slots	
Description: Doctor needs to click on Manage Time Slots tab before removing a time slot	
Actors: Doctor	
Trigger: This event will be triggered when the user clicks on the Remove Time Slot button	
Normal Flow: 1. The Doctor will click on the Remove Time Slot button 2. System prompts the user with a Confirm/Cancel Remove Time Slot 3. The Doctor will click on the Confirm button 4. System will validate the time slot and remove it from their schedule 5. Doctor can view the updated schedule	
Sub-Flows: None	
Alternative/Exceptional Flows: 4a. System is unable to remove the time slot 5a. Doctor is unable to view the updated schedule	

Name: Log Out	ID: D09
Stakeholders and goals: Doctor – Doctor can successfully log out from the system	
Description: Doctor wants to log out from the system.	
Actors: Doctor	
Trigger: This event will be triggered when the user clicks on the Log Out button	
Normal Flow: 1. The Doctor will click on the Log Out button 2. Doctor successfully logged out from the system	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

3.3 Use Case Diagram - Hospital Staff



3.3.1 User Stories - Hospital Staff

- As a hospital staff I want to be able to log in so that I can access the system
- As a hospital staff I want to be able to log out so that I can exit the system
- As a hospital staff I want to be able to modify the account information so that the data is up to date
- As a hospital staff I want to be able to view appointments so that I can monitor the appointments
- As a hospital staff I want to send reminders to the patients so that the patients do not forget their appointment
- As a hospital staff I want to be able to send calendar invites to the patients so that I can notify the patients about the upcoming events
- As a hospital staff I want to be able to view the doctor's appointment details so that I can know what time the appointment and the patients' information
- As a hospital staff I want to manage the time slots for the facilities of the hospital so that I can notify the patients about the available time slots
- As a hospital staff I want to be able to view the details of other appointment types so that I can know what time the appointment and it's information
- As a hospital staff I want to be able to cancel the different appointment types so that patients are notified about the cancellation.
- As a hospital staff I want to be able to reschedule the different appointment types so that patients are notified about the rescheduling.
- As a hospital staff I want to be able to upload information about Cost and Financing so that the patients can understand what specialties are available
- As a hospital staff I want to be able to modify information about Cost and Financing so that the information is up to date
- As a hospital staff I want to be able to delete information about Cost and Financing so that the information is up to date
- As a hospital staff I want to be able to upload information about Facilities and Services so that the patients can understand what specialties are available
- As a hospital staff I want to be able to modify information about Facilities and Services so that the information is up to date
- As a hospital staff I want to be able to delete information about Facilities and Services so that the information is up to date

3.3.2 Use Case Description - Hospital Staff

Name: Log In (Hospital Staff)	**ID: H01**
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully login into the system	
Description: A Hospital Staff needs to login before using the system.	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the Login button	
Normal Flow: 1. **The Hospital Staff will click on the Login button** 2. **System shows the login form** 3. **The Hospital Staff will enter the login credentials** 4. **System will validate the Hospital Staff** 5. **Hospital Staff will be logged in**	
Sub-Flows: None	
Alternative/Exceptional Flows: **4a. System is unable to authenticate user** **5a. Hospital Staff is unable to log in to the system**	

Name: Update Account Information	ID: H02
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully update their account information	
Description: A Hospital Staff should be able to update their account information on the system.	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the Update Account Information button	
Normal Flow: 1. The Hospital Staff will click on the Update Account Information button 2. System shows the Update Account Information form 3. Hospital Staff enters the information they want to update 4. The Hospital Staff will click on the Update button 5. System validates the updated information 6. Hospital Staff's Account Information is updated	
Sub-Flows: None	
Alternative/Exceptional Flows: **5a. Hospital Staff's information is not valid** **6a. Failed to update Hospital Staff's Account Information**	

Name: View Doctor Appointment	ID: H03
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully view the doctor appointments made by patients	
Description: Hospital Staff wants to view the doctor appointment made by the patients.	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the View 'Doctor Appointments' button	
Normal Flow: 1. The Hospital Staff will click on the View 'Doctor Appointments' button 2. System will display a list of doctor appointments made by the patients	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: View Service Appointment	ID: H04
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully view the service appointments made by patients	
Description: Hospital Staff wants to view the service appointment made by the patients.	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the View 'Service Appointments' button	
Normal Flow: 1. **The Hospital Staff will click on the View 'Service Appointments' button** 2. **System will display a list of service appointments made by the patients**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Reschedule Service Appointment	ID: H05
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully reschedule the service appointments made by patients	
Description: Hospital Staff wants to reschedule the service appointment made by the patients.	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the View 'Reschedule' button	
Normal Flow: 1. **The Hospital Staff will click on the 'Reschedule' button** 2. **System will display a page with the original appointment booking details and an input field for the user to choose date** 3. **The Hospital Staff will choose a date** 4. **System validates the date and displays the available time slots for that day** 5. **The Hospital Staff will choose the time slot** 6. **System validates the time slot and updates the appointment details inside the database and notifies the patient about it.**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** Staff chooses a date when the service is not available	

Name: Cancel Service Appointment	ID : H06
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully cancel the service appointment	
Description: Hospital Staff wants to cancel the service appointment	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the View 'Cancel' button	
Normal Flow: 1. **The Hospital Staff will click on the 'Cancel' button** 2. **System will display a modal to ask for confirmation** 3. **The Hospital Staff clicks 'Confirm'** 4. **System cancels the appointment and sends an email to the relevant patient**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** Staff cancels deletion	

Name: Send Reminders	ID: H07
Stakeholders and goals: Hospital Staff – Hospital Staff can send a reminder to the patients regarding their appointment	
Description: Hospital Staff should be able to send reminders to the patients.	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the Send Reminders button	
Normal Flow: 1. **The Hospital Staff will click on the Send Reminders button** 2. **System will send a reminder to the patient**	
Sub-Flows: None	
Alternative/Exceptional Flows: **2a. System failed to send reminder to the patient** **2b. System will display error message in the UI**	

Name: Send Calendar Invites	ID: H08
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully send calendar invites to the patients	
Description: Hospital Staff should be able to send calendar invites to the patients.	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the Send Calendar Invites button	
Normal Flow: 1. **The Hospital Staff will click on the Send Calendar Invites button** 2. **System will display the calendar invites form** 3. **Hospital Staff fill the information in the form and click the submit button** 4. **System validates the input** 5. **System sends calendar invites to the patients**	
Sub-Flows: None	
Alternative/Exceptional Flows: **4a. Hospital Staff input is invalid** **5a. System unable to send calendar invites** **5b. System will display error message in the UI**	

Name: Manage Facilities	ID: H09
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully manage the facilities offered by a hospital	
Description: Hospital Staff wants to manage the facilities offered by a hospital	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the Manage Facilities Button	
Normal Flow: 1. **The Hospital Staff will click on the Manage Facilities button** 2. **System will display the list of facilities offered by the hospital along with three more actions to perform, Add Facilities, Modify Facilities and Delete Facilities** 3. **Hospital staff chooses an action to perform** 4. **System will redirect the user to the relevant page / display a modal for the user to perform the action**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Add New Facilities	ID: H10
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully add new facilities	
Description: Hospital Staff wants to add new facilities	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the 'Add Facilities' button	
Normal Flow: 1. **The Hospital Staff will click on the 'Add Facilities' button** 2. **System will display a form with fields to create a new facility** 3. **Hospital Staff enters the relevant information inside the form** 4. **System validates the information entered and inserts the new record inside the database**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** Hospital Staff entered invalid information	

Name: Modify Facilities	ID : H11
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully modify an existing facility	
Description: Hospital Staff wants to modify existing facility	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the 'Modify' button	
Normal Flow: 1. **The Hospital Staff will click on the 'Modify' button** 2. **System will display a form with fields containing existing data and allowing the staff to edit** 3. **Hospital Staff enters the most updated valid information inside the fields** 4. **System validates the information entered and updates the existing record inside the database**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** Hospital Staff entered invalid information	

Name: Delete Facilities	ID : H12
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully delete an existing facility	
Description: Hospital Staff wants to delete existing facility	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the 'Delete' button	
Normal Flow: 1. **The Hospital Staff will click on the 'Delete' button** 2. **System will display a modal to confirm deletion** 3. **Hospital Staff clicks on 'Confirm' button** 4. **System removes the facility from the screen and deletes the row of record from the database**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** Hospital Staff cancels deletion and clicks on 'Cancel' button	

Name: Manage Cost	ID: H13
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully manage the costs included by a hospital	
Description: Hospital Staff wants to manage the costs included by a hospital	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the Manage Costs Button	
Normal Flow: 1. **The Hospital Staff will click on the Manage Cost button** 2. **System will display the list of costs included by the hospital along with three more actions to perform, Add Cost, Modify Cost and Delete Cost** 3. **Hospital staff chooses an action to perform** 4. **System will redirect the user to the relevant page / display a modal for the user to perform the action**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Add New Cost	ID: H14
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully add a new cost	
Description: Hospital Staff wants to add new cost	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the 'Add Cost' button	
Normal Flow: 1. **The Hospital Staff will click on the 'Add Cost' button** 2. **System will display a form with fields to create a new cost** 3. **Hospital Staff enters the relevant information inside the form** 4. **System validates the information entered and inserts the new record inside the database**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** Hospital Staff entered invalid information	

Name: Modify Cost	ID : H15
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully modify an existing cost	
Description: Hospital Staff wants to modify existing cost	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the 'Modify' button	
Normal Flow: 1. **The Hospital Staff will click on the 'Modify' button** 2. **System will display a form with fields containing existing data and allowing the staff to edit** 3. **Hospital Staff enters the most updated valid information inside the fields** 4. **System validates the information entered and updates the existing record inside the database**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** Hospital Staff entered invalid information	

Name: Delete Cost	ID : H16
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully delete an existing cost	
Description: Hospital Staff wants to delete existing cost	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the 'Delete' button	
Normal Flow: 1. **The Hospital Staff will click on the 'Delete' button** 2. **System will display a modal to confirm deletion** 3. **Hospital Staff clicks on 'Confirm' button** 4. **System removes the facility from the screen and deletes the row of record from the database**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** Hospital Staff cancels deletion and clicks on 'Cancel' button	

Name: Manage Services	ID: H17
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully manage the services offered by a hospital	
Description: Hospital Staff wants to manage the services offered by a hospital	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the 'Manage Services' Button	
Normal Flow: 1. **The Hospital Staff will click on the Manage Services button** 2. **System will display the list of services offered by the hospital along with three more actions to perform, Add New Services, Modify Services and Delete Services** 3. **Hospital staff chooses an action to perform** 4. **System will redirect the user to the relevant page / display a modal for the user to perform the action**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Add New Services	ID: H18
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully add a new service offered by the hospital	
Description: Hospital Staff wants to add new service offered by the hospital	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the 'Add Service' button	
Normal Flow: 1. **The Hospital Staff will click on the 'Add Service' button** 2. **System will display a form with fields to create a new service** 3. **Hospital Staff enters the relevant information inside the form** 4. **System validates the information entered and inserts the new record inside the database**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** Hospital Staff entered invalid information	

Name: Modify Services	ID : H19
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully modify an existing service offered by the hospital	
Description: Hospital Staff wants to modify existing service offered by the hospital	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the 'Modify' button	
Normal Flow: 1. **The Hospital Staff will click on the 'Modify' button** 2. **System will display a form with fields containing existing data and allowing the staff to edit** 3. **Hospital Staff enters the most updated valid information inside the fields** 4. **System validates the information entered and updates the existing record inside the database**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** Hospital Staff entered invalid information	

Name: Delete Service	ID : H20
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully delete an existing service	
Description: Hospital Staff wants to delete existing service	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the 'Delete' button	
Normal Flow: 1. **The Hospital Staff will click on the 'Delete' button** 2. **System will display a modal to confirm deletion** 3. **Hospital Staff clicks on 'Confirm' button** 4. **System removes the facility from the screen and deletes the row of record from the database**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** Hospital Staff cancels deletion and clicks on 'Cancel' button	

Name: Manage Time Slots about Facilities and Services	ID:H21
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully manage the time slots for facilities and services	
Description: Hospital Staff wants to manage the time slots for facilities and services	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the Modify Time Slots button	
Normal Flow: 1. **The Hospital Staff will check on facilities** 2. **Hospital then will then click on the Modify Time Slots button** 3. **System will display the Modify Time Slots page** 4. **Hospital staff entered the updated information** 5. **System validates the input** 6. **Time Slots updated**	
Sub-Flows: None	
Alternative/Exceptional Flows: **4a. Information entered is invalid** **5a. Failed to update Time Slots**	

Name: Reschedule Appointment	ID: H22
Stakeholders and goals: Hospital Staff can successfully re-schedule an appointment	
Description: Hospital Staff wants to reschedule an appointment.	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the Reschedule button	
Normal Flow: 1. **The Hospital Staff will click on the Reschedule button** 2. **System shows the reschedule appointment form** 3. **Hospital Staff enters a new date and time for the appointment** 4. **System validates the information entered by the Hospital Staff** 5. **Appointment rescheduled** 6. **Patient notified**	
Sub-Flows: None	
Alternative/Exceptional Flows: **4a. Information entered by the Hospital Staff is invalid** **5a. Failed to reschedule an appointment**	

Name: Cancel Appointment	ID:H23
Stakeholders and goals: Hospital Staff can successfully cancel an appointment	
Description: Hospital Staff wants to cancel an appointment.	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the Cancel button	
Normal Flow: 1. **The patient will click on the Cancel button** 2. **System prompts the user for confirmation on the cancellation** 3. **Hospital Staff confirms the cancellation** 4. **Appointment cancelled** 5. **Patients will be notified.**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a. Users cancel the appointment cancellation** **4a. Failed to cancel an appointment**	

Name: View Appointment Details	ID: H24
Stakeholders and goals: Hospital Staff can successfully view the appointment details	
Description: Hospital Staff wants to view the appointment details.	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the View Details	
Normal Flow: 1. **The Hospital Staff will click on the View Details button** 2. **System will display the detailed version of the appointment on the page**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Log Out	ID:H25
Stakeholders and goals: Hospital Staff – Hospital Staff can successfully log out from the system	
Description: Hospital Staff wants to log out from the system.	
Actors: Hospital Staff	
Trigger: This event will be triggered when the user clicks on the Log Out button	
Normal Flow: 1. **The Hospital Staff will click on the Log Out button** 2. **Hospital Staff successfully logged out from the system**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

3.4 Use Case Diagram - Hospital Admin

3.4.1 User Stories – Hospital Admin

- As a hospital admin I want to be able to log in so that I can access the system
- As a hospital admin I want to be able to create a hospital staff account so that I can assign some task to the staff
- As a hospital admin I want to be able to create a doctor account so that the patient can choose the doctor
- As a hospital admin I want to be able to manage the medical specialties so that the doctor can be assigned to a specialties
- As a hospital admin I want to be able to delete the medical specialties so it can be removed if it is not relevant to the hospital anymore

3.4.2 Use Case Description – Hospital Admin

Name: Log In (Hospital Admin)	ID: HA01
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully login into the system	
Description: A Hospital Admin needs to login before using the system.	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Login button	
Normal Flow: 1. **The Hospital Admin will click on the Login button** 2. **System shows the login form using a modal** 3. **The Hospital Admin will enter the login credentials** 4. **System will validate the Hospital Admin credentials** 5. **Hospital Admin will be logged in**	
Sub-Flows: None	
Alternative/Exceptional Flows: **4a. System is unable to authenticate user** **5a. Hospital Admin is unable to log in to the system**	

Name: Manage Doctors	ID: HA02
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully manage doctors and their information	
Description: A Hospital Admin should be able to manage doctors and doctors' information.	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Manage Doctors button	
Normal Flow: 1. **The Hospital Admin will click on the Manage Doctors button** 2. **System shows the Manage Doctors page which contains the list of doctors and options to Create Doctor Account, Modify Doctor Account, Delete Doctor Account** 3. **Hospital Admin choose on one of the button** 4. **System records the input from Hospital Staff** 5. **Doctors data is updated**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

Name: Create Doctor Account	ID: HA03
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully add a create new doctor account	
Description: Hospital Admin wants to add a new doctor into the system.	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Create Doctor Account button	
Normal Flow: 1. **The Hospital Admin will click on the Create Doctor Account button** 2. **System shows the create new doctor form** 3. **Hospital Admin enters doctor information** 4. **System validates the information entered by the Hospital Admin** 5. **New doctor account added into the system**	
Sub-Flows: None	
Alternative/Exceptional Flows: **4a. Information entered by the Hospital Admin is invalid** **5a. Failed to add a new doctor account into the system**	

Name: Modify Doctor Account Password	ID: HA04
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully modify the doctor account password.	
Description: Hospital Admin wants to edit the doctor account password.	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Modify Password button	
Normal Flow: 1. **The Hospital Admin will click on the Modify Password button** 2. **System shows the modify doctor password form** 3. **Hospital Staff enters the password matching the validation** 4. **System validates the password entered by the Hospital Admin** 5. **Doctor Account Password updated**	
Sub-Flows: None	
Alternative/Exceptional Flows: **4a. Password entered by the Hospital Admin does not validate** **5a. Failed to update doctor's information**	

Name: Delete Doctor Account	ID: HA05
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully delete a doctor from the system	
Description: Hospital Admin wants to remove a doctor from the system.	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Delete Doctor button	
Normal Flow: 1. **The Hospital Admin will click on the Delete Doctor button** 2. **System prompts the user for confirmation on the deletion using a modal** 3. **Hospital Admin confirms the deletion** 4. **Doctor deleted from the system**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a. Hospital Admin cancels the deletion** **4a. Failed to delete the doctor**	

Name: Create Hospital Staff Account	ID: HA06
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully create an account for the hospital staff	
Description: Hospital Admin should be able to create a Hospital Staff Account	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Add Hospital Staff button	
Normal Flow: 1. **The Hospital Admin will click on the Add Hospital Staff button** 2. **System shows the create Hospital Staff Form** 3. **Hospital Admin enters the information for the account including an initial password** 4. **The Hospital Admin will click on the Create button** 5. **System validates the information** 6. **Hospital Staff Account created**	
Sub-Flows: None	
Alternative/Exceptional Flows: **5a. Hospital Admin entered invalid information** **6a. Failed to create hospital staff account**	

Name: Modify Hospital Staff Account's Password	ID: HA07
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully modify the Hospital Staff Account Password	
Description: Hospital Admin should be able to change Hospital Staff Account Password.	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Modify Password button	
Normal Flow: 1. **The Hospital Admin will click on the Modify Password button** 2. **System displays a change password form** 3. **Hospital Admin fills in a valid password** 4. **Password updated successfully**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a. Hospital Admin entered an invalid password**	

Name: Delete Hospital Staff Account	ID : HA08
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully delete a Hospital Staff account	
Description: Hospital Admin wants to delete a Hospital Staff account.	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Delete Staff button	
Normal Flow: 1. **The Hospital Admin will click on the Delete Staff button** 2. **System prompts the user for confirmation using a modal** 3. **Hospital Admin confirms the deletion of the account** 4. **Account deleted**	
Sub-Flows: None	
Alternative/Exceptional Flows: **2a. Admin cancels the deletion** **4a. Failed to delete account**	

Name: Manage Medical Specialties	ID: HA09
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully manage the medical specialties offered by a hospital	
Description: Hospital Admin wants to manage medical specialties about a hospital.	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Manage Medical Specialties button	
Normal Flow: 1. **The Hospital Admin will click on the Manage Specialties button** 2. **System will display the Manage Specialties page** 3. **Hospital can select either of the actions related to specialties** 4. **System will redirect the user to the relevant page to perform the action or generate a pop-up screen**	
Sub-Flows: None	
Alternative / Exceptional Flows : None	

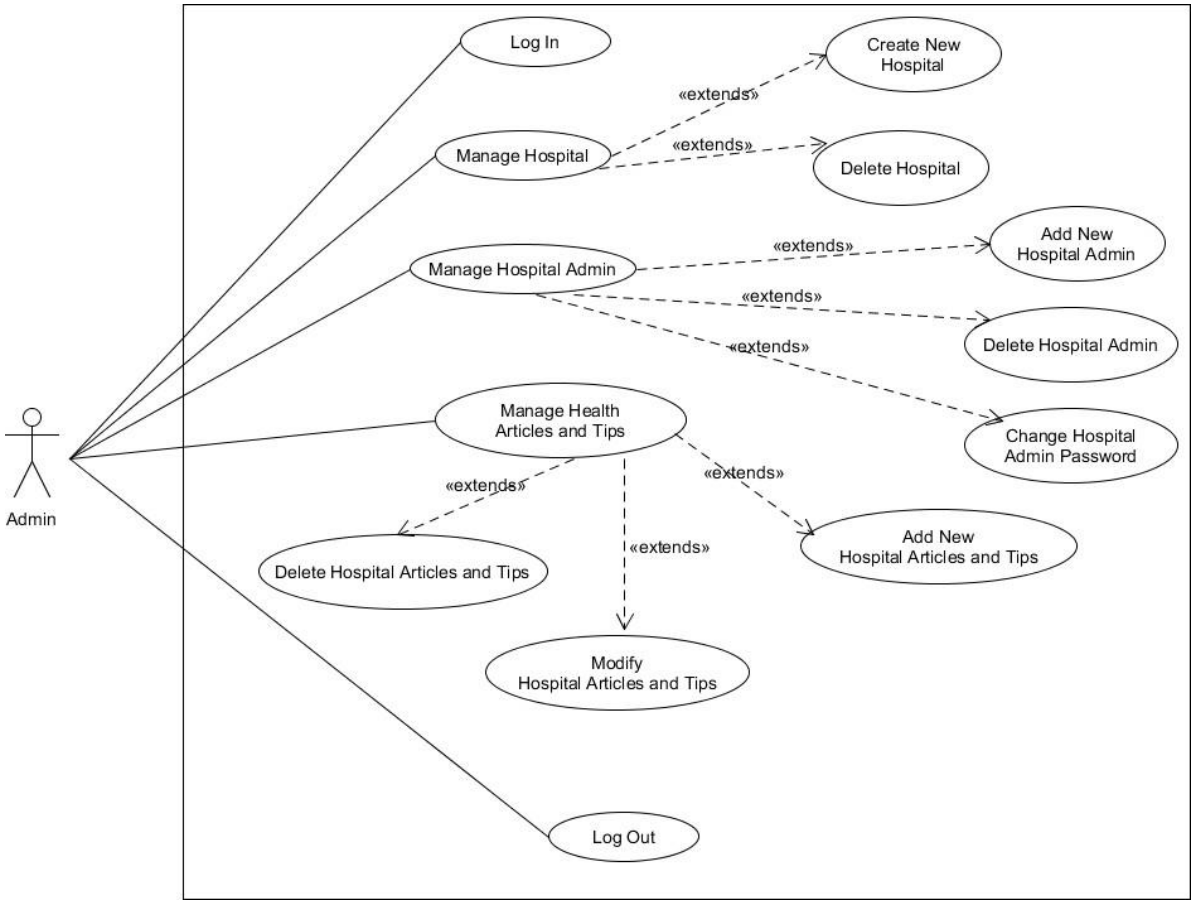
Name: Add New Medical Specialties	ID: HA10
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully add new specialties of the hospital	
Description: Hospital Admin wants to add a new specialty of the hospital	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Add Medical Specialties	
Normal Flow: 1. **The Hospital Admin will click on the Add Medical Specialties button** 2. **System will display a form to create a Medical Specialty** 3. **The Hospital Admin can enter relevant information about the specialty** 5. **System validates the information entered and inserts it into the database**	
Sub-Flows: None	
Alternative / Exceptional Flows : 3a. The user enters invalid information	

Name: Modify Specialties	ID: HA11
Stakeholders and goals: Hospital Admin – Hospital Admin can modify the specialties information	
Description: Hospital Admin wants to modify an existing specialty	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Modify Medical Specialties	
Normal Flow: 1. **The Hospital Admin will click on the Modify Specialties button** 2. **System will display the previously entered information which is editable** 3. **The Hospital Admin will enter the relevant updated information** 4. **System validates the information and updates it in the database**	
Sub-Flows: None	
Alternative / Exceptional Flows : 3a. The user enters invalid information	

Name: Delete Medical Specialties	ID : HA12
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully delete medical specialties	
Description: Hospital Admin wants to delete a Medical Specialty	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Delete button	
Normal Flow: 1. **The Hospital Admin will click on the Delete button** 2. **System prompts the user for confirmation using a modal** 3. **Hospital Admin confirms the deletion of the specialty** 4. **Specialty deleted**	
Sub-Flows: None	
Alternative/Exceptional Flows: **2a. Admin cancels the deletion** **4a. Failed to delete account**	

Name: Log Out	ID : HA13
Stakeholders and goals: Hospital Admin – Hospital Admin can successfully log out from the system	
Description: Hospital Admin wants to log out from the system.	
Actors: Hospital Admin	
Trigger: This event will be triggered when the user clicks on the Log Out button	
Normal Flow: 1. **The Hospital Admin will click on the Log Out button** 2. **Hospital Admin successfully logged out from the system**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

3.5 Use Case Diagram – Admin



3.5.1 User Stories - Admin

- As an admin I want to be able to log in so that I can access the system
- As an admin I want to be able to log out so that I can exit the system
- As an admin I want to be able to create a hospital so that the hospital admin, staff, and doctor can access the system
- As an admin I want to be able to delete the account the hospital so that the database does not store any irrelevant information.
- As an admin I want to be able to modify the health care articles and tips so that the patients are aware of the latest updates and innovations made in the health industry.
- As an admin I want to be able to upload the health care articles and tips so that the patients can be made more health conscious.
- As an admin I want to be able to remove the health care articles and tips so that the patients do not read obsolete health care information.
- As an admin I want to be able to upload FAQ's so that the patients are able to clear their doubts on the most frequently asked questions.
- As an admin I want to be able to edit FAQ's so that the patients are aware of any changes made to the answers to the FAQ's.
- As an admin I want to be able to delete FAQ's so that the patients do believe information that has become obsolete.
- As an admin I want to be able to create a hospital admin account so that the hospital can be accessed and managed
- As an admin I want to be able to delete a hospital admin account so that it will be removed from the database

3.5.2 Use Case Description - Admin

Name: Log In (Admin)	**ID: A01**
Stakeholders and goals: Admin – Admin can successfully login into the system	
Description: Admin needs to login before using the system.	
Actors: Admin	
Trigger: This event will be triggered when the user clicks on the Login button	
Normal Flow: 1. **The Admin will click on the Login button** 2. **System shows the login form** 3. **The Admin will enter the login credentials** 4. **System will validate the Admin** 5. **Admin will be logged in**	
Sub-Flows: None	
Alternative/Exceptional Flows: **4a. System is unable to authenticate user** **5a. Admin is unable to log in to the system**	

Name: Create Hospital Admin Account	ID: A02
Stakeholders and goals: Admin – Admin can successfully create an account for the hospital admin	
Description: Admin should be able to create a hospital account for the hospital admin.	
Actors: Admin	
Trigger: This event will be triggered when the user clicks on the Create Hospital Admin Account button	
Normal Flow: 1. **The Admin will click on the Create Hospital Admin Account button** 2. **System shows the create hospital admin account form** 3. **Admin enters the information for the account including an initial password** 4. **The Admin will click on the Create button** 5. **System validates the information** 6. **Hospital Admin Account created**	
Sub-Flows: None	
Alternative/Exceptional Flows: **5a. Admin's information is not valid** **6a. Failed to create hospital admin account**	

Name: Delete Hospital Admin Account	ID: A03
Stakeholders and goals: Admin – Admin can successfully delete a hospital admin account	
Description: Admin wants to delete a hospital admin account.	
Actors: Admin	
Trigger: This event will be triggered when the user clicks on the Delete Hospital Admin Account button	
Normal Flow: 1. **The Admin will click on the Delete Hospital Admin Account button** 2. **System prompts the user for confirmation** 3. **Admin confirms the deletion of the account** 4. **Account deleted**	
Sub-Flows: None	
Alternative/Exceptional Flows: **5a. Admin cancel the deletion** **6a. Failed to delete account**	

Name: Modify Hospital Admin Password	ID : A04
Stakeholders and goals: Admin – Admin can modify hospital admin password	
Description: Admin wants to modify hospital admin password	
Actors: Admin	
Trigger: This event will be triggered when the user clicks on the Modify Password button	
Normal Flow: 1. **The Admin will click on the Modify Password button** 2. **System displays a change password form** 3. **Admin enters valid format password** 4. **System validates the password and updates it in the Firebase**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a. Admin enters invalid password**	

Name: Create Hospital	ID : A05
Stakeholders and goals: Admin – Admin can successfully create a hospital	
Description: Admin should be able to create a hospital	
Actors: Admin	
Trigger: This event will be triggered when the user clicks on the Create Hospital Button	
Normal Flow: 1. The Admin will click on the Create Hospital Button 2. System shows the create hospital form 3. Admin enters the valid information for the hospital 4. The Admin will click on the Create button 5. System validates the information 6. Hospital Admin Account created	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a. Admin entered invalid information** **6a. Failed to create hospital admin account**	

Name: Delete Hospital	ID : A06
Stakeholders and goals: Admin – Admin can successfully delete a hospital	
Description: Admin should be able to delete a hospital	
Actors: Admin	
Trigger: This event will be triggered when the user clicks on the Delete Hospital button	
Normal Flow: 1. The Admin will click on the Delete Hospital Button 2. System prompts the user for confirmation using a modal 3. Admin confirms the deletion of the hospital 4. Hospital deleted	
Sub-Flows: None	
Alternative/Exceptional Flows: **2a. Admin cancels the deletion** **4a. Failed to delete hospital record**	

Name: Manage Health Articles and Tips	ID: A07
Stakeholders and goals: Admin – Admin can successfully manage the information about the health articles and tips	
Description: Admin wants to manage information about the health articles and tips.	
Actors: Admin	
Trigger: This event will be triggered when the user clicks on the Manage Health Articles and Tips button	
Normal Flow: 1. The Admin will click on the Manage Health Articles and Tips button 2. System will display a page containing several articles and tips with several options offered to add, modify and delete. 3. Admin clicks on the relevant action button 4. System redirects the user to the action page for the user to perform	
Sub-Flows: None	
Alternative/Exceptional Flows:	

Name: Add Health Articles and Tips	ID: A08
Stakeholders and goals: Admin – Admin can successfully add a new health article / tip	
Description: Admin wants to add new health article / tip	
Actors: Admin	
Trigger: This event will be triggered when the user clicks on the Add Articles / Add Tips button	
Normal Flow: **1. The Admin will click on the Add Articles / Add Tips button** **2. System will display a page with fields to create Articles or Tips** **3. The Admin will enter relevant information into the fields** **4. System validates the information entered**	
Sub-Flows: None	
Alternative/Exceptional Flows:	

Name: Modify Health Articles and Tips	ID: A09
Stakeholders and goals: Admin – Admin can successfully modify an existing health article or tip	
Description: Admin wants to modify an existing health article or tip	
Actors: Admin	
Trigger: This event will be triggered when the user clicks on the Modify button	
Normal Flow: **1. The Admin will click on the Modify button** **2. System will display a form with existing values and allow the user to edit it** **3. The Admin will click on the Save button** **4. System validates the input and updates it inside the database**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** The Admin enters invalid information **3b.** The Admin clicks on the Cancel button	

Name: Delete Health Articles and Tips	ID : A10
Stakeholders and goals: Admin – Admin can successfully delete an existing health article or tip	
Description: Admin wants to delete an existing health article or tip	
Actors: Admin	
Trigger: This event will be triggered when the user clicks on the Delete button	
Normal Flow: **1. The Admin will click on the Delete button** **2. System prompts the user for confirmation of the deletion** **3. The Admin confirms deletion** **4. Systems deletes it from the database**	
Sub-Flows: None	
Alternative/Exceptional Flows: **3a.** The Admin cancels deletion	

Name: Log Out	ID: A11
Stakeholders and goals: Admin – Admin can successfully log out from the system	
Description: Admin wants to log out from the system.	
Actors: Admin	
Trigger: This event will be triggered when the user clicks on the Log Out button	
Normal Flow: **1. The Admin will click on the Log Out button** **2. Admin successfully logged out from the system**	
Sub-Flows: None	
Alternative/Exceptional Flows: None	

4. Non-functional Requirements

4.1 Security Requirements

Security should ensure to provide confidentiality, integrity, and availability/authenticity. To provide confidentiality, all sensitive data such as passwords will be encrypted in the database. Integrity will be provided by using an integrity check and data validation functions to ensure that the data has not been modified. Login will be required prior to performing any actions by all users to provide availability/authenticity.

4.2 Reliability Requirements

Reliability shows how well the website is able to perform without having any failures and errors. With this, we must ensure that there will be minimal bugs in our system. Also, our website should be able to handle a certain amount of workload during usage.

4.3 Performance Requirements

Performance requirements define how well the software system accomplishes certain functions under specific conditions. In this case, the website should respond quickly as soon as the user navigates around the website. It should take no more than 5 seconds to load every page that the user navigates into. Additionally, the website should be able to be accessed by 400 users at the same time.

4.4 Maintainability Requirements

Maintainability defines the ease of fixing bugs and faults which are found within the software. Our code should allow new features and modifications to be implemented without hindering other functions of the software. This means the code should be neat whereby the admin/programmer can easily understand the structure and make respective changes.

4.5 Scalability Requirements

Scalability is the degree to which the site is able to expand its processing capabilities to support business growth. For instance, currently our site is only viable for scheduling appointments for the different categories of patients based in Singapore. However, our team might inculcate more functionalities to the site in the future to gain higher customer loyalty. One of the features anticipated by our team includes collaborating with different insurance companies so that we can also offer the patients different policies and grants at a lower rate, thus helping them tackle the growing inflation rate. Besides, we also hope to expand our influence across the neighbouring South East Asian countries which would be specifically catered to meet their needs and demands.

4.6 Usability Requirements

Usability shows how easily the user is able to operate and interact with the site. Our team hopes to achieve high user satisfaction, for which we aim to make our site easily accessible to all, user-friendly with an appealing interface and easily navigable without the need to refer to a manual. Since we aspire to cover a large group of audience, we would try to make our website as secure and reliable as possible, in order to gain high customer loyalty.